

Power Function & Types of Error, Pt. 2 (Revised)

CSU, Chico Math 351

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Calculating β

Our previous example involved a one-sided hypothesis test. Depending on our alternative hypothesis, the probability of making a Type II error will differ, and our calculation for β will change.

Calculating β

Let μ_0 denote the value of μ in the null hypothesis and let μ_a denote the actual value of μ . Let $\beta(\mu_a)$ be the probability of a Type II error if the actual value of the mean is μ_a .

For a one tailed test $H_1 : \mu > \mu_0$, the value of β at μ_a is

$$P\left(Z < z_{1-\alpha}^* - \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right)$$

and for $H_1 : \mu < \mu_0$:

$$P\left(Z > z_{\alpha}^* + \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right)$$

For a two-tailed test we get:

$$P\left(Z < z_{1-\alpha/2}^* - \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right)$$

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Definition of Power

The **power** of a test is the probability that we reject a false null hypothesis ($1-\beta$). The plot of power versus the value of μ is called the **power curve**. Ideally, one should design tests that have large power.

While we cannot directly choose power as we can our level of significance α , the following affect power:

- ▶ Choice of α (increasing it increases the power)
- ▶ Population Standard deviation (decreasing it increases the power)
- ▶ **Sample size** (increasing it increases the power)

Determining Sample Size

Since sample size is the easiest of these to modify, we can decide what sample size is needed to achieve a certain level of power for a given value of μ .

Determining Sample Size: Example

Say an experimenter wishes to test $H_0 : \mu = 100$ vs. $H_1 : \mu > 100$ with $\alpha = 0.05$, and wants $1 - \beta = 0.6$ when $\mu_a = 103$. What is the smallest sample size that will achieve this? Assume a normal distribution with $\sigma = 14$.

We will need two equations for $\bar{X}^*$: One for H_0 's distribution and once for H_1 's distribution.

$$P(Z \geq \frac{\bar{X}^* - 100}{14/\sqrt{n}}) = 0.05$$

so

$$\frac{\bar{X}^* - 100}{14/\sqrt{n}} = 1.6449$$

and

$$\bar{X}^* = 100 + 1.6449 * \frac{14}{\sqrt{n}}$$

Determining Sample Size: Example

The power for $\mu = 103$ would then be:

$$1 - \beta = P(Z > \frac{\bar{X}^* - 103}{14/\sqrt{n}}) = 0.60$$

and using RStudio:

$$qnorm(1 - 0.60) = -0.2533$$

so

$$\frac{\bar{X}^* - 103}{14/\sqrt{n}} = -0.2533$$

and

$$\bar{X}^* = 103 - 0.2533 * \frac{14}{\sqrt{n}}$$

Determining Sample Size: Example

Using our two equations:

$$100 + 1.6449 * \frac{14}{\sqrt{n}} = 103 - 0.2533 * \frac{14}{\sqrt{n}}$$

$$(1.6449 + 0.2533) * \left(\frac{14}{\sqrt{n}}\right) = 3$$

$$\frac{14}{1.5804} = \sqrt{n}$$

$$n = (14/1.5804)^2 \approx 78.47$$

To ensure a power of at least 0.60 at $\mu = 103$, you would need at least **79** observations (round up, since 78 observations are insufficient to reach 0.60 power)

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Exercise: Parking Tickets

A city manager audits the parking tickets issued by city parking officers to determine the number of tickets that were contested by the car owner and found to be improperly issued. In past years, the number of improperly issued tickets per officer had a normal distribution with mean $\mu = 380$ and $\sigma = 35.2$. Because there has recently been a change in the city's parking regulations, the city manager suspects that the mean number of improperly issued tickets has increased.

An audit of 50 randomly selected officers is conducted to test whether there has been an increase in improper tickets. Use the sample data test the research hypothesis that the mean number of improperly issued tickets is greater than 380. The audit generates the following data: $n = 50$ and $\bar{X} = 390$. Test at significance level $\alpha = 0.01$ whether the mean number of improperly issues tickets is **greater than 380**.

Example: Parking Tickets

Given from the previous slide:

$$\mu_0 = 380$$

$$\bar{X} = 390$$

$$\sigma = 35.2$$

$$\alpha = 0.01$$

$$n = 50$$

$$z_{1-\alpha}^* = qnorm(1 - \alpha) = 2.3263$$

Exercise: Parking Tickets

- ▶ Establish hypotheses: null (H_0) and alternative (H_1).
- ▶ Calculate test statistics and p-value
- ▶ State conclusion (reject H_0 ? Context of problem?)
- ▶ Based on conclusion, if an error was made, was it Type I or Type II?

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References

references I

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